TH550/THP550

Industrial Robot

Manual for TH500 and THP550 Differences

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Shibaura Machine

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Introduction

The THP550 has been designed specifically for high-speed transfer of lightweight objects and features a lightweight and low-inertia arm unit for enabling agility in the 0.2-second cycle time range combined with high durability capable of 24-hour continuous operation at 120 cycles per minute.
[Applications]
The THP550 is ideal for applications involving high-speed continuous transfer between conveyors and box packing such as for crystalline silicon solar cells, foodstuffs, pharmaceuticals, and other items.

This manual describes the differences from the TH550 industrial robot, which has the same arm length.
This manual is divided into the sections below.

Chapter 1: Specifications

Chapter 2: Robot Height and Z-axis Position

Chapter 3: Permissible Load Conditions

Chapter 4: Cables Between Robot and Controller

Chapter 5: CN0 Layout and Number of Inputs/Outputs

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1. Specifications

No	Name		TH550	**THP550**	Remarks
No	Item		Specification		
1	Type		Horizontal multi-articullation		
2	No. of controlled axes		4 axes		
3	Arm length	Full length	550 (mm)		
		Arm 1	300 (mm)		
		Arm 2	250 (mm)		
4	Working envelope	Axis 1	±120 (deg)		
		Axis 2	±145 (deg)		
		Axis 3	0 to 150 (mm)		
		Axis 4	±360 (deg)		
5	Maximum speed	Axis 1	375 (deg/sec)		
		Axis 2	600 (deg/sec)		
		Axis 3	2,000 (mm/sec)		
		Axis 4	2,000 (deg/sec)		
		Composite	6.21 (m/sec)		
6	Payload mass	Rated	2 (kg)	**1 (kg)**	*1
		Max	5 (kg)	**2 (kg)**	
7	Standard cycle time (when transferring 1 kg)		0.33 (sec)	**0.29 (sec)**	*2
8	Permissible load inertia		0.05 (kg•m^2)	**0.01 (kg•m^2)**	3
9	Positioning repeatability	X–Y	±0.01 (mm)	**±0.015 (mm)**	*4
		Z (axis 3)	±0.01(mm)		
		C	±0.02 (deg)	**±0.02 (deg)**	
10	Drive system		AC servo motor for all axes		
11	Robot body	Mass	29 kg	**26 kg**	
12	Hand wires and pipes		5 inputs and 4 outputs 4mm diameter x 4	**8 inputs and 8 outputs 4mm diameter x 4**	
13	Options		• Ceiling suspension model • Z-axis long stroke • Top cap model	• **Ceiling suspension model** • **Z-axis long stroke** • **Top cap model** • **IP** • **Cleanroom model**	*5

*1 : In the TH550, the rated transportable weight is 2 kg, and the maximum transportable weight is 5 kg.
In the **THP550**, the rated transportable weight is 1 kg, and the maximum transportable weight is 2 kg.

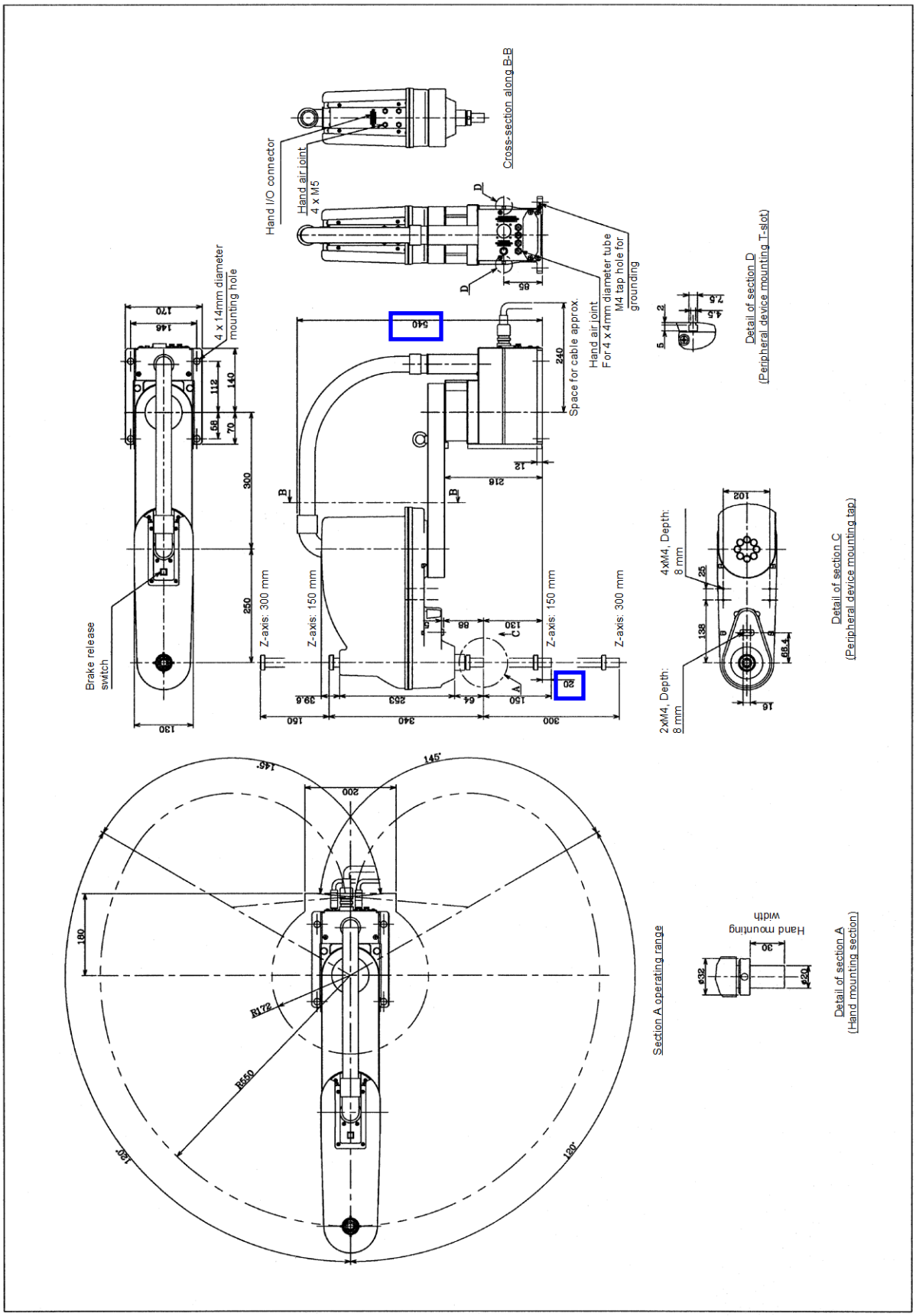
*2 : In the TH550, the cycle time is based on a rated weight of 2 kg. An overload error can occur if continuous operation is performed.
The TH550 is suitable for continuous operation on workpieces of 2 to 5 kg where the robot takes time to stop.
In the **THP550**, the cycle time is based on a rated weight of 1 kg. It is capable of 24-hour continuous operation at 120 cycles per minute.
The **THP550** is suitable for continuous operation on lightweight workpieces of 1 to 2 kg without stopping.

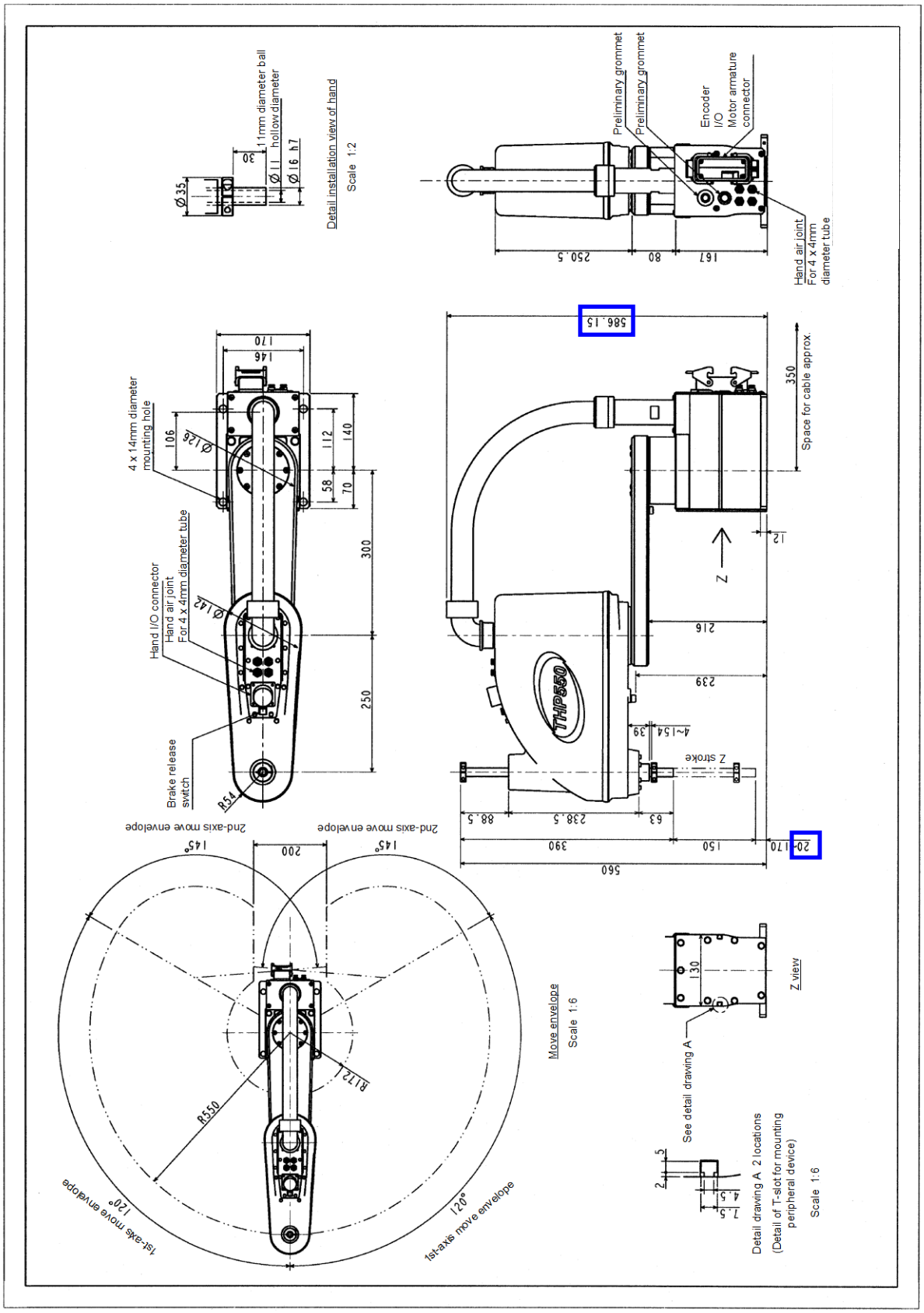
*3 : Because the THP550 has a smaller maximum transportable weight than the TH550, the permissible load inertia is also smaller.

*4 : The THP550 is a robot whose primary purpose is high-speed continuous operation, and so its positioning repeatability accuracy is lower than the TH550.

*5 : For information about the availability of other options, please contact the Sales Department.

2. Robot Height and Z-axis Position

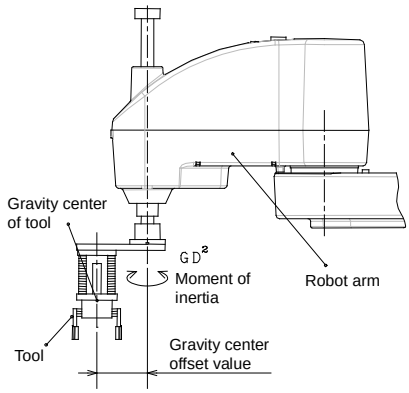




The TH550 has a total height of 540 mm. The **THP550** has a total height of 586.15 mm, which is higher than the TH550.
Also, the Z-axis origin position is -20 mm from the robot mounting surface in the TH550, but it is +20 mm in the **THP550**.

3. Permissible Load Conditions

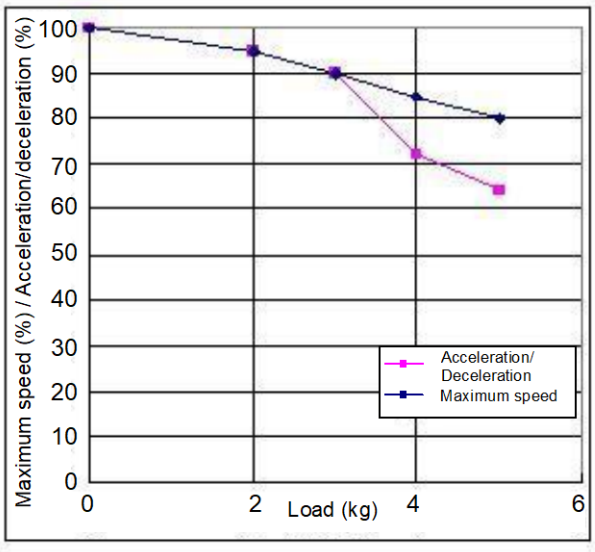
The differences between the permissible load conditions in the TH550 and THP550 are shown below.



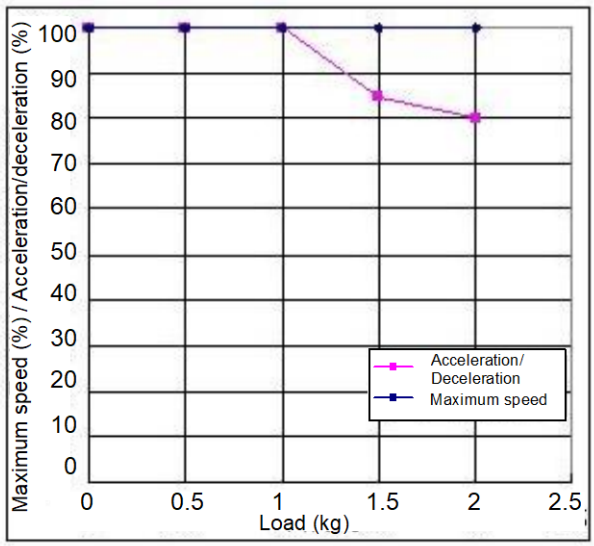
Permissible load conditions

Model	TH550	**THP550**
Conditions	Allowance	**Allowance**
Mass	Rated 2kg (Max. 5kg) Maximum acceleration setting at 2 kg or less	**Rated 1 kg (Max. 2 kg)** **Maximum acceleration setting at 1 kg or less**
Moment of inertia	Max 0.05 (kg•m²)	**Max 0.01 (kg•m²)**
Gravity center offset	Max 100mm (load≤10 kg)	**Max 50 mm** **(load≤2 kg)**

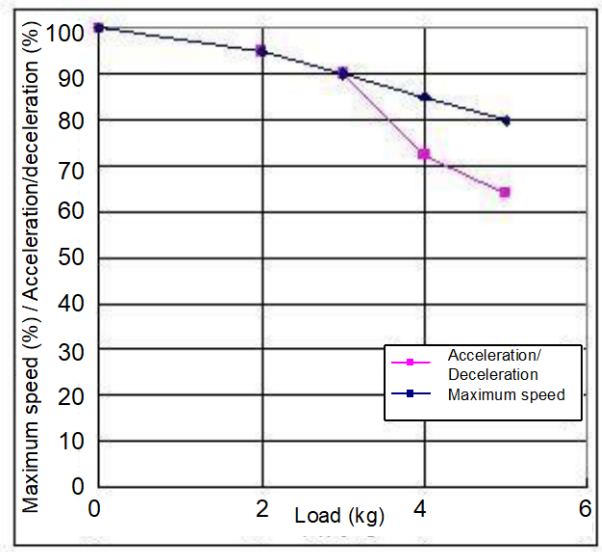
a) If there is no offset in the load, the acceleration is limited by the load mass. The figures below show the permissible load conditions when there is no offset.



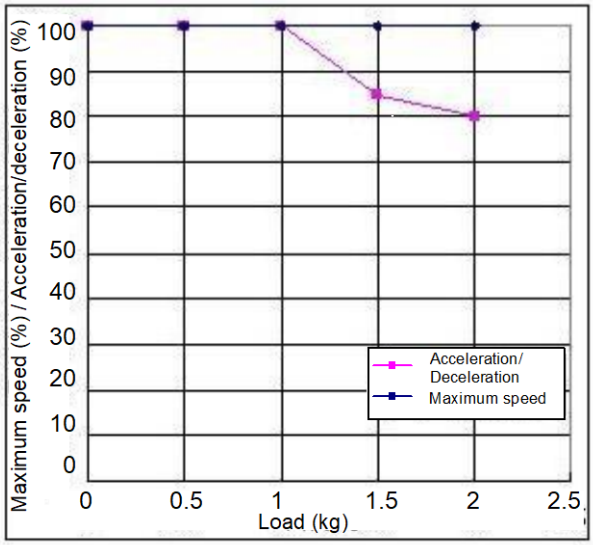
TH550: Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



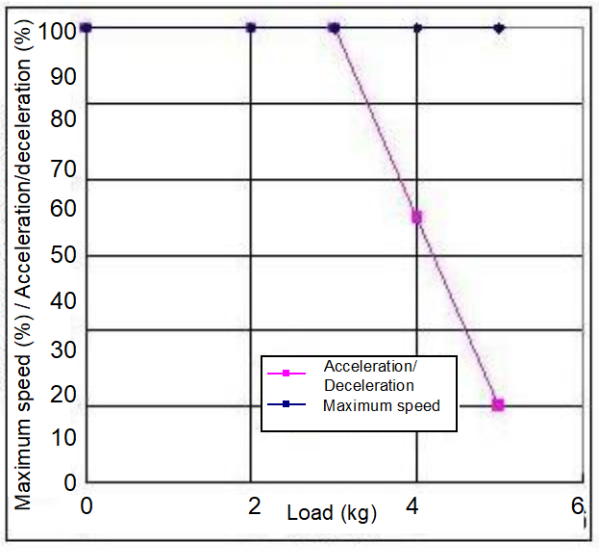
THP550: Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



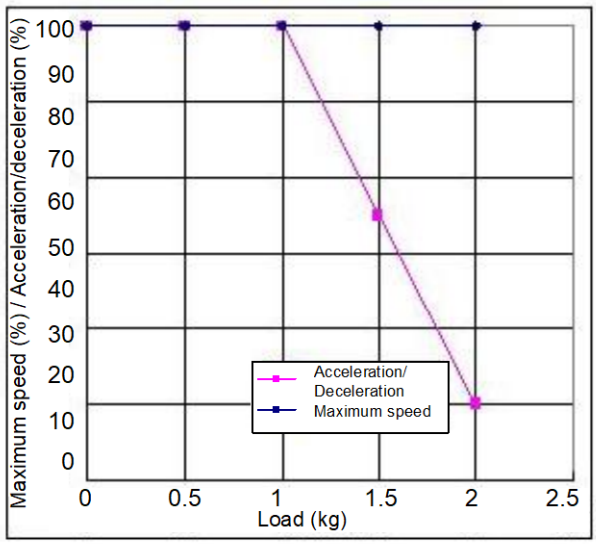
TH550: Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)



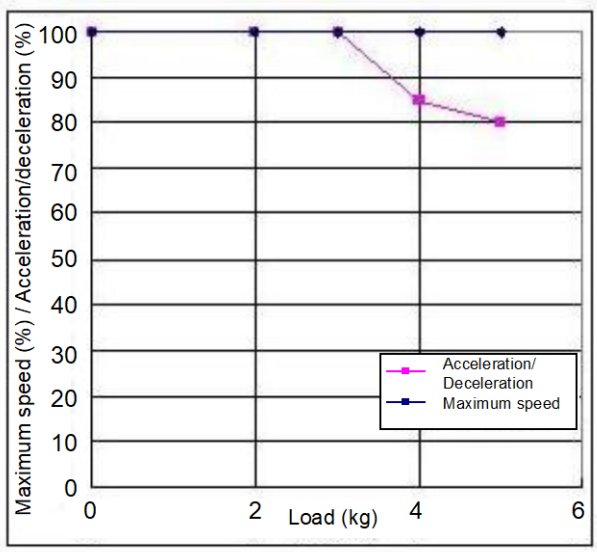
THP550: Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)



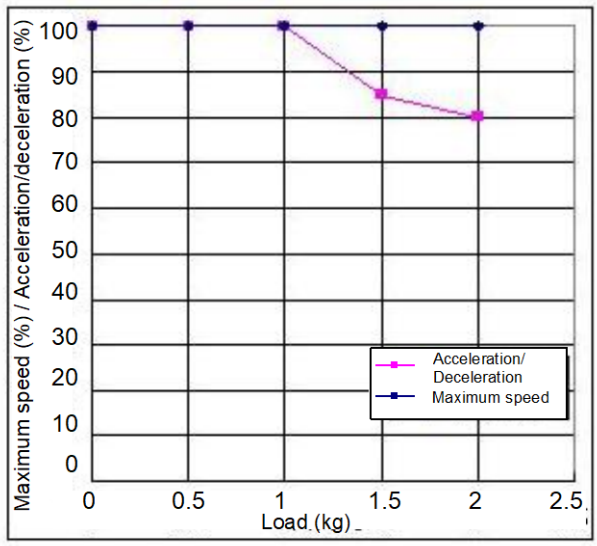
TH550: Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)



THP550: Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)

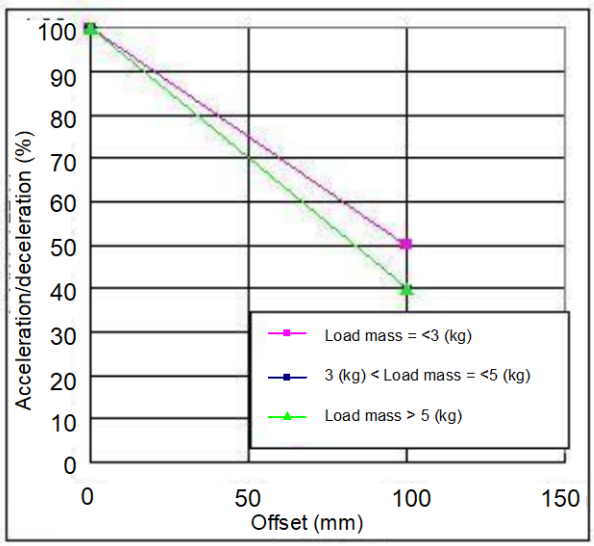


TH550: Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

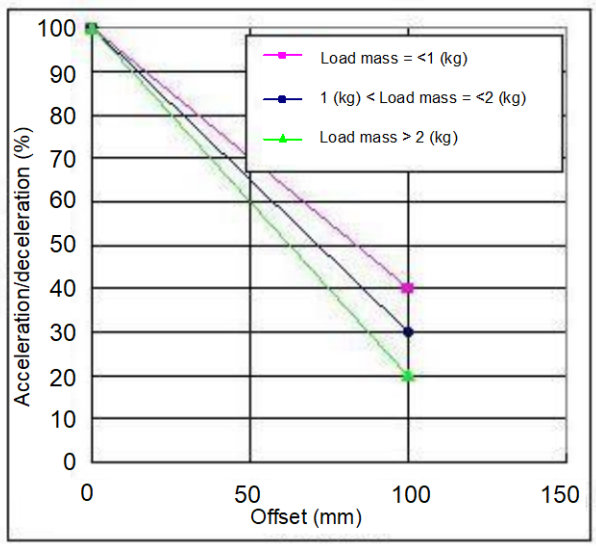


THP550: Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

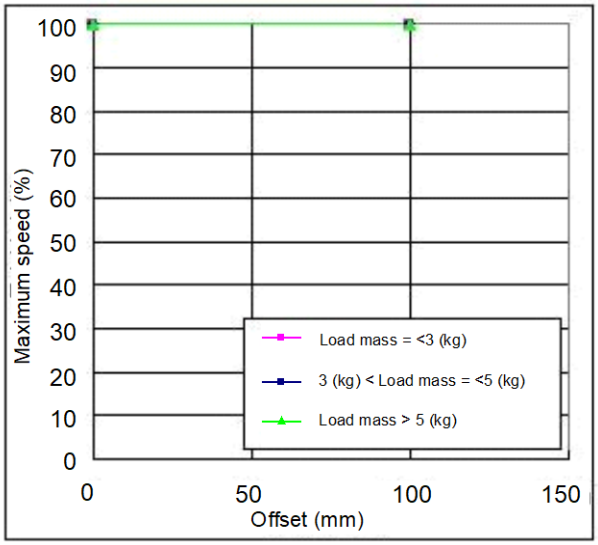
b) If there is offset in the load, the acceleration that is set by the load mass in item a) above is further limited by the offset value. (Axis 3 is not limited by the offset.) The figures below show the rate of decrease (axes 1, 2, and 4) of the acceleration based on the offset value.



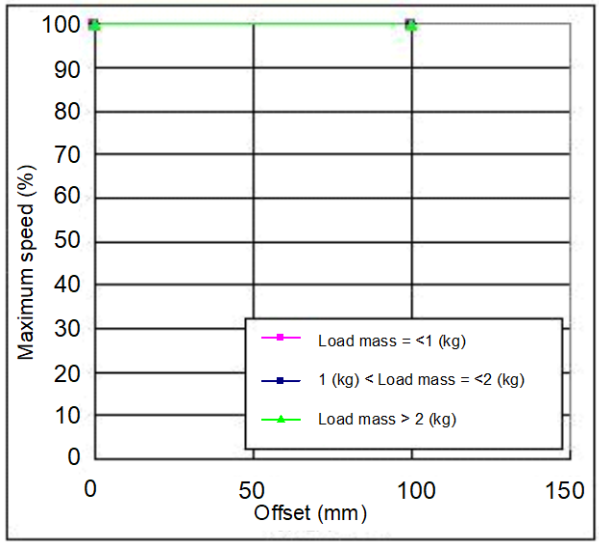
TH550: Setting of acceleration/deceleration time in relation to offset value (Axis 1)



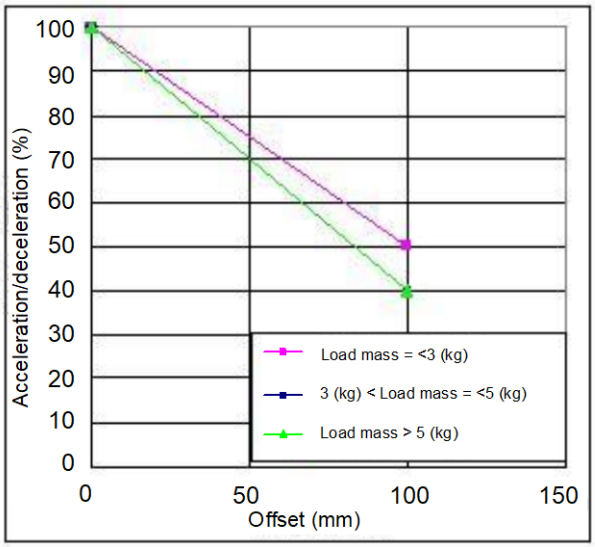
THP550: Setting of acceleration/deceleration time in relation to offset value (Axis 1)



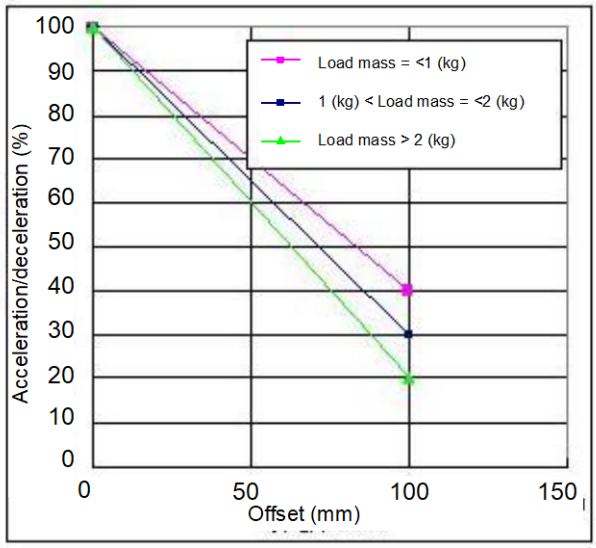
TH550: Setting of maximum speed in relation to offset value (Axis 1)



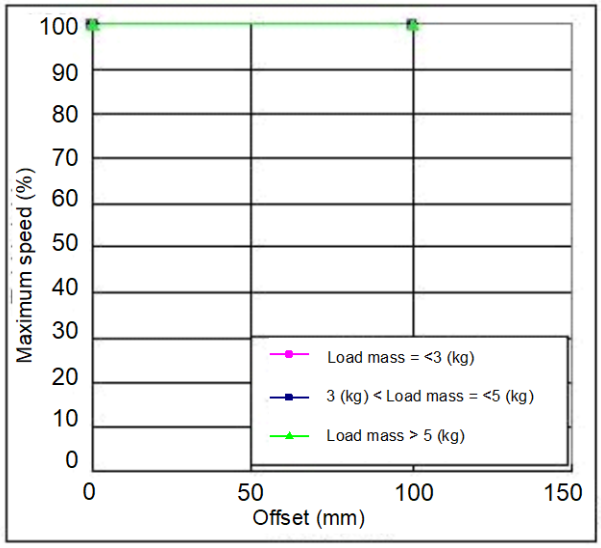
THP550: Setting of maximum speed in relation to offset value (Axis 1)



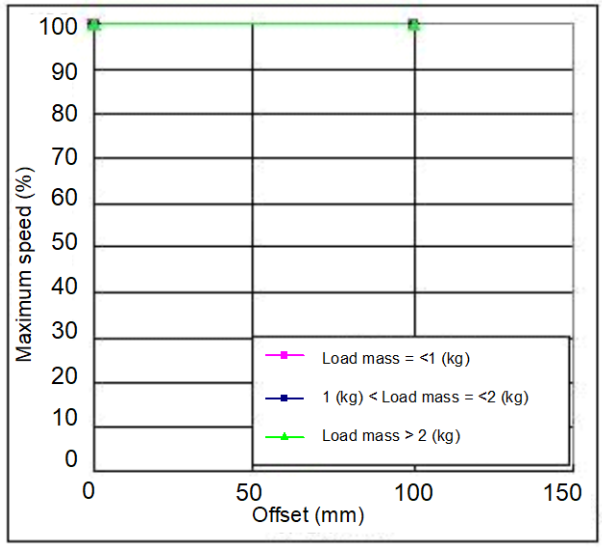
TH550: Setting of acceleration/deceleration time in relation to offset value (Axis 2)



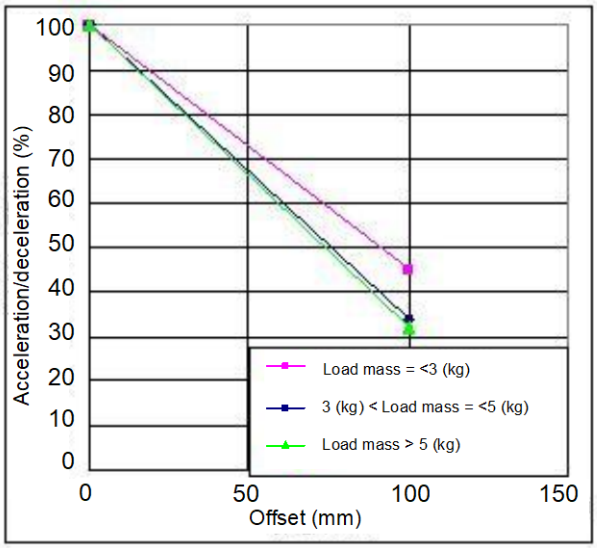
THP550: Setting of acceleration/deceleration time in relation to offset value (Axis 2)



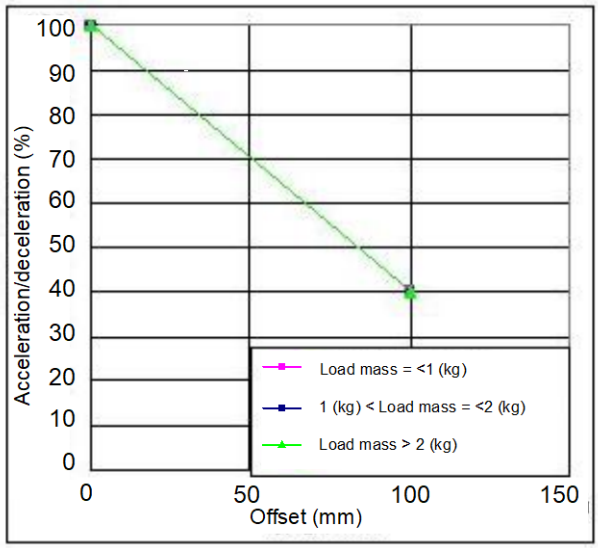
TH550: Setting of maximum speed in relation to offset value (Axis 2)



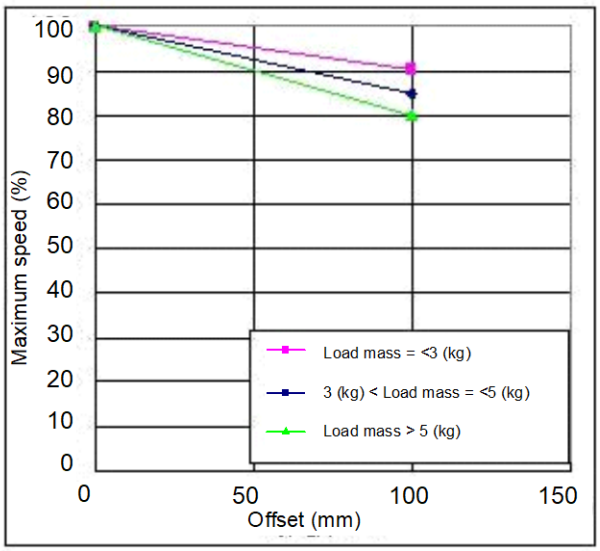
THP550: Setting of maximum speed in relation to offset value (Axis 2)



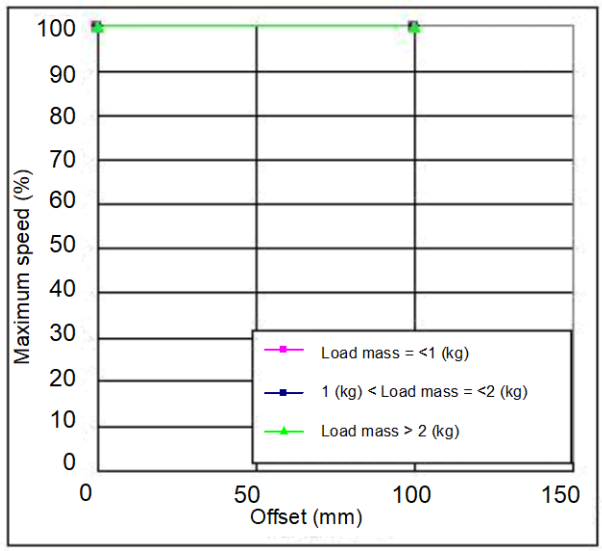
TH550: Setting of acceleration/deceleration time in relation to offset value (Axis 4)



THP550: Setting of acceleration/deceleration time in relation to offset value (Axis 4)



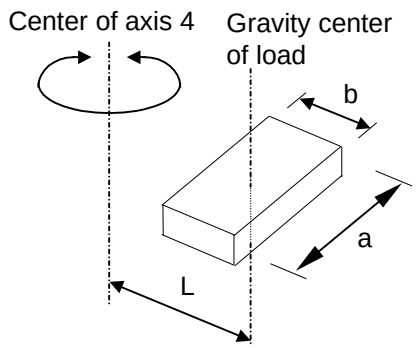
TH550: Setting of maximum speed in relation to offset value (Axis 4)



THP550: Setting of maximum speed in relation to offset value (Axis 4)

c) Moment of inertia

Shown below is a model simplifying the robot and load, and arithmetic expression of moment of inertia of load.



L : Distance from axis 4 center to gravity center of load (m)

a : Width of load (m)

b : Length of load (m)

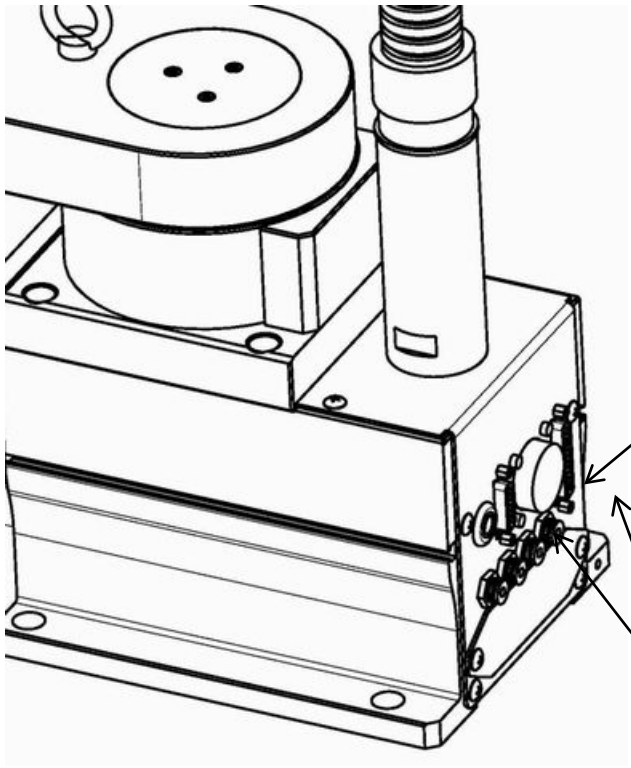
M : Mass of load (kg)

Moment of inertia ($kg \bullet m^2$)

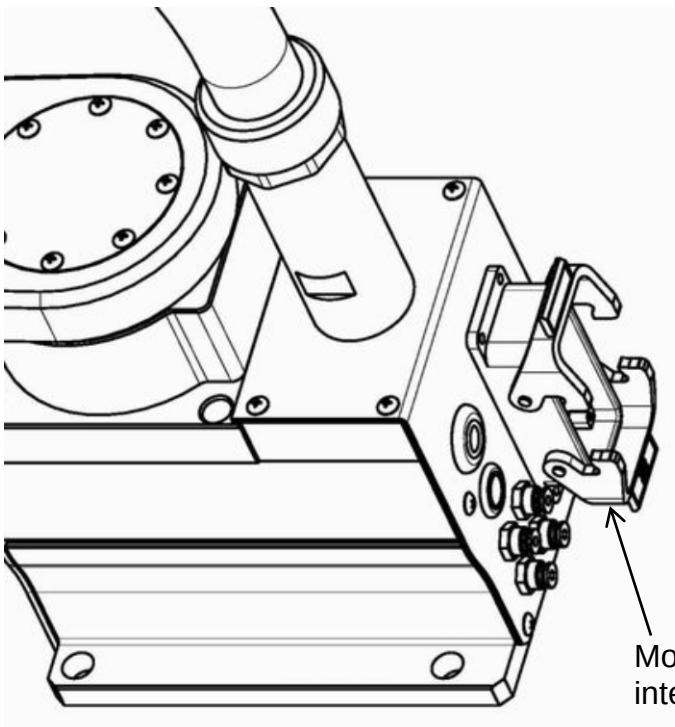
$$= \frac{M}{12}(a^2 + b^2) + ML^2$$

4. Cables Between Robot and Controller

The TH550 uses one Cannon connector and two D-Sub connectors on the robot side.
The **THP550** uses a single integrated connector.



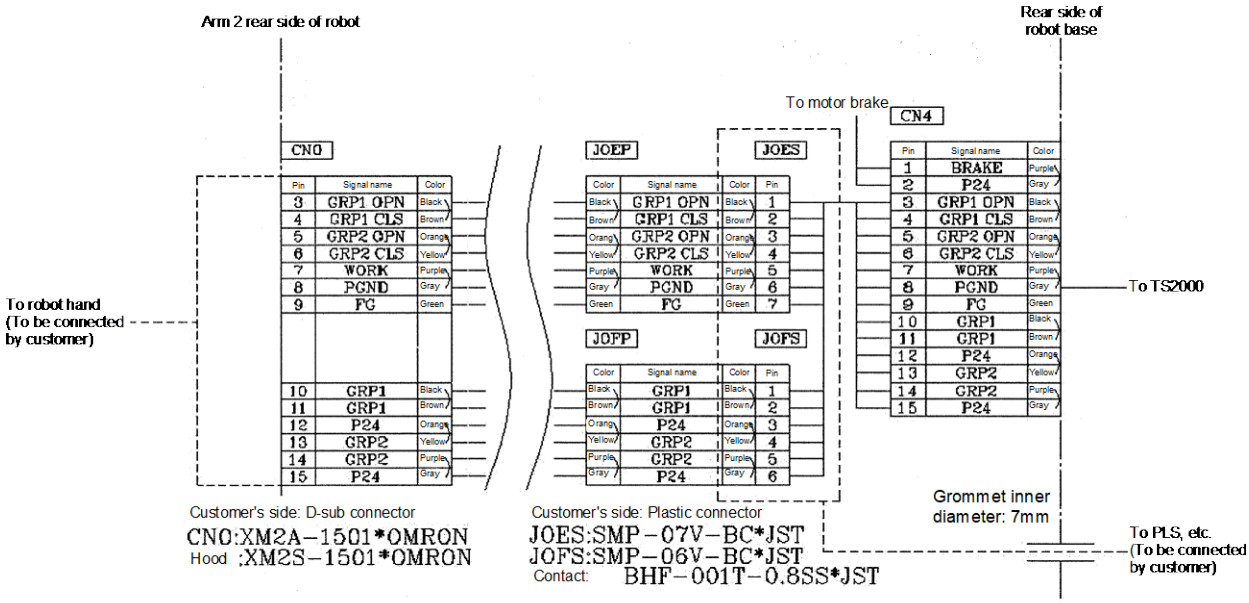
TH550



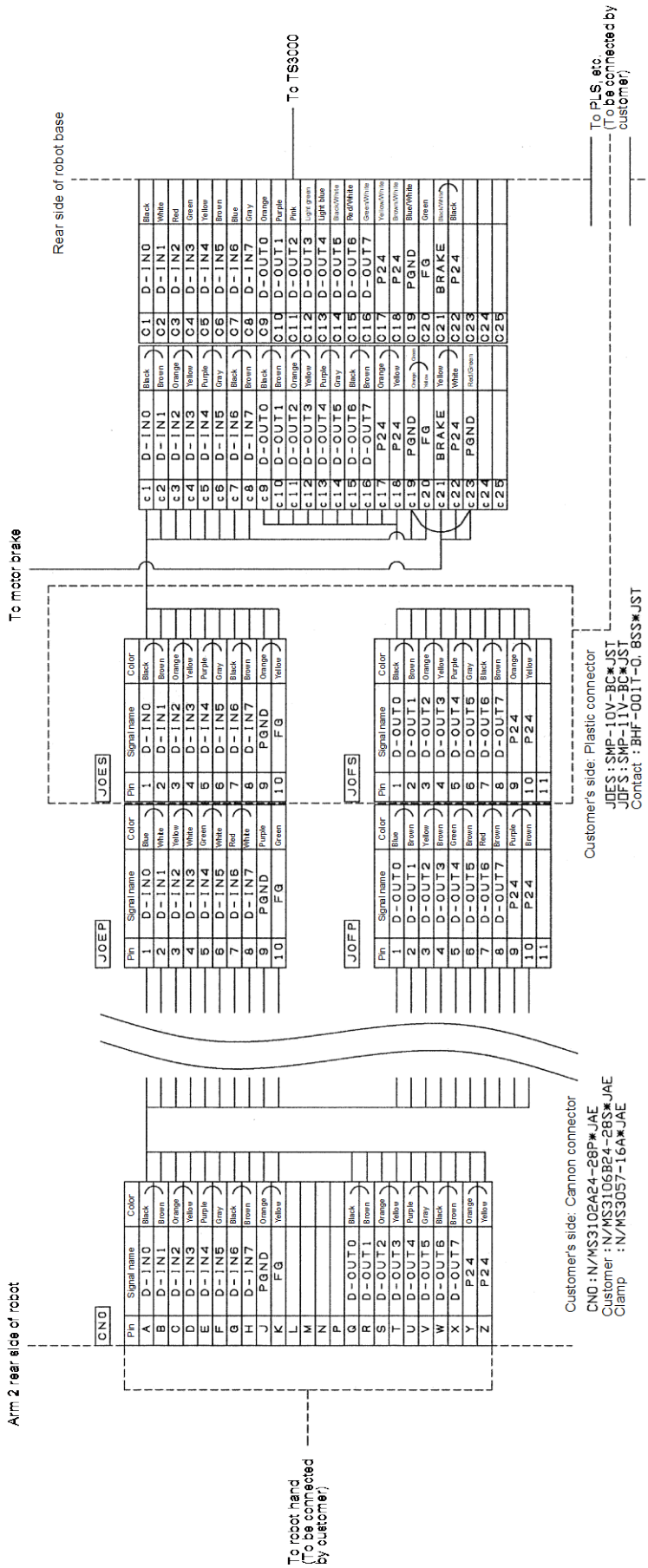
THP550

5. CN0 Layout and Number of Inputs/Outputs

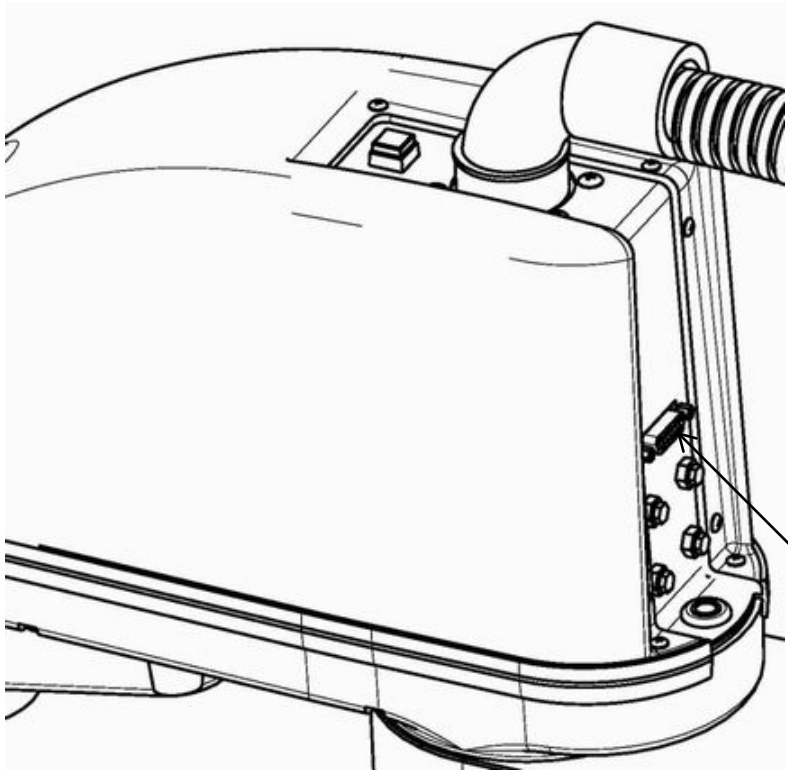
The TH550 is provided with 5 inputs and 4 outputs for the CN0 (D-Sub) connector.
The **THP550** is provided with 8 inputs and 8 outputs for the CN0 (cannon) connector.
The connector position may vary.



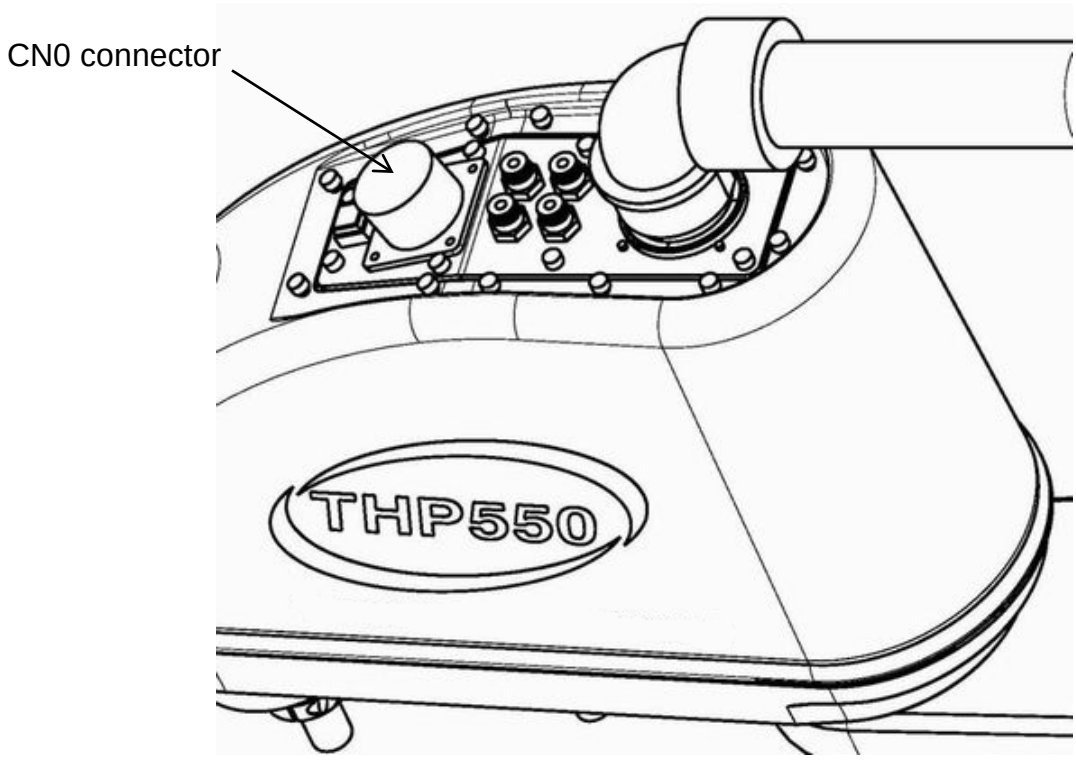
TH550 Tool Wire



THP550 Tool Wire



TH550



THP550